

DERIVING PRO DISTRIBUTIVE

Silippo Bonchi
University of Pisa

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GRAM LOGICS MONOIDAL CAT

Elena Di Lavore Mario Román

Tallinn University of Technology

FROM EGORIES

Sam Staton
University of Oxford

IMPERATIVE CATEGORIES

= distributive copy-discard categories with a
Denotational semantics of effectful imperative

COMMANDS SYNTAX

$C ::= \text{skip}$
 $| \text{abort}$

$| C_1; C_2$

$| \text{if } b \text{ then } c_1 \text{ else } c_2$

SEMANTICS

$$(\otimes) \quad \begin{array}{c} x \text{ --- } x \\ \vdots \\ x \text{ --- } x \end{array}$$

$$(+ \quad x^N \text{ --- } \text{---} x^N$$

$$(\otimes) \quad \begin{array}{c} x \text{ --- } \boxed{C_1} \text{ --- } \boxed{C_2} \text{ --- } x \\ \vdots \quad \quad \quad \vdots \\ x \text{ --- } \boxed{C_1} \text{ --- } \boxed{C_2} \text{ --- } x \end{array}$$

$$(+ \quad \begin{array}{c} x^N \text{ --- } \boxed{b} \text{ --- } \left(\begin{array}{c} \boxed{C_1} \\ \boxed{C_2} \end{array} \right) \text{ --- } x^N \end{array}$$

compatible and uniform coproduct-trace

programs

GUARDS SYNTAX

$b ::= L$

$| R$

$| \neg b$

$| b(e_1, \dots, e_k)$

SEMANTICS

(+) $x^N \text{---} \bullet \text{---} 1$
 $\vdash 1$

(+) $\vdash 1$
 $x^N \text{---} \bullet \text{---} 1$

(+) $x^N \text{---} \boxed{b} \text{---} \text{---} 1$
 $\text{---} 1$

($\otimes$) $x \text{---} \vdots \text{---} \bullet \text{---} \vdots \text{---} \bullet \text{---} x$
 $\vdots \text{---} \vdots \text{---} \vdots \text{---} \vdots$
 $e_1 \text{---} \vdots \text{---} \vdots \text{---} \vdots \text{---} \vdots$
 $e_k \text{---} \vdots \text{---} \vdots \text{---} \vdots \text{---} \vdots$
 $\vdots \text{---} \vdots \text{---} \vdots \text{---} \vdots$
 $b \text{---} 1+1$

DERIVING PROGRAM LOGICS

GUARDED MONOID

- $(C, 1, \cdot, 0)$ monoid with absorbing element
- $(B, (-+_L, -): B^3 \rightarrow B, \neg(-): B \rightarrow B, L \in B)$ such that
$$a+_L b = a$$
$$L+_a \neg L = a$$
$$a+_b c = c+_b a$$
$$\neg(a+_b c) = \neg a+_b \neg c$$
$$(a+_b c)+_d (e+_b f) = (a+_d e)+_b (c+_d f)$$
$$(a+_b c)+_d (a+_e c) = a+_d (b+_e c) c$$
- an operation $(-\oplus, -): C \times B \times C \rightarrow C$ such that
$$f+_L g = f$$
$$f+_b g = g+_b f$$
$$(f+_b g) \cdot h = (f \cdot h)+_b (g \cdot h)$$
$$(f+_b g)+_a (h+_b l) = (f+_a h)+_b (g+_a l)$$
$$(f+_a g)+_b (f+_c g) = f+_b (a+_c) g$$

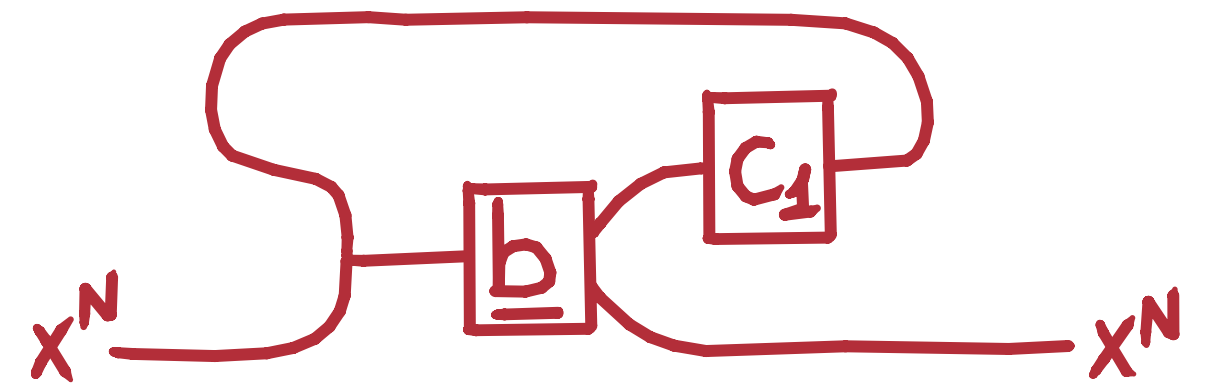
| while b do c

| assert p

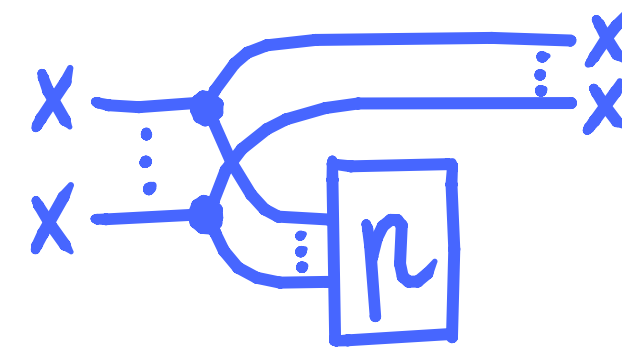
| $x_i \leftarrow e$

| $(x_{i_1}, \dots, x_{i_k}) \leftarrow \Delta$

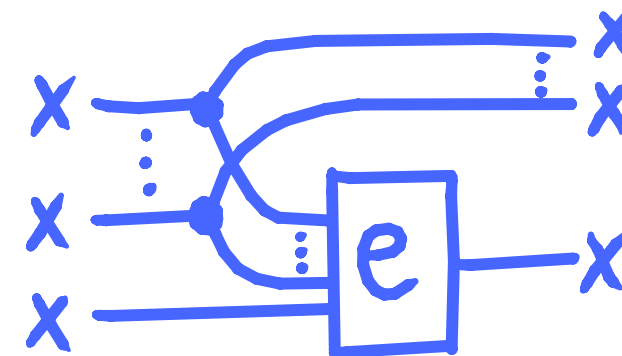
(+)



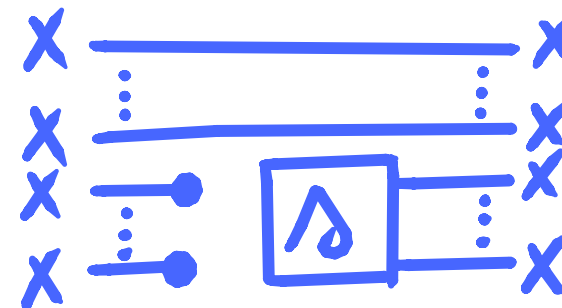
($\otimes$)



($\otimes$)



($\otimes$)



PROGRAM TRIPLES

STATE $(P, Q: 1 \rightarrow X^N)$

PREDICATE $(P, Q: X^N \rightarrow 1)$

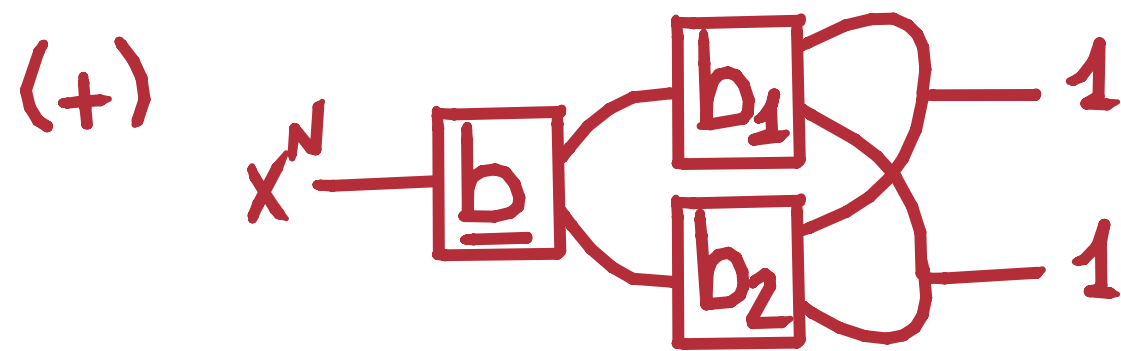
CORRECTNESS $\llbracket P \rrbracket; \llbracket c \rrbracket \leq \llbracket Q \rrbracket$

$\llbracket P \rrbracket \leq \llbracket c \rrbracket; \llbracket Q \rrbracket$

INCORRECTNESS $\llbracket P \rrbracket; \llbracket c \rrbracket \geq \llbracket Q \rrbracket$

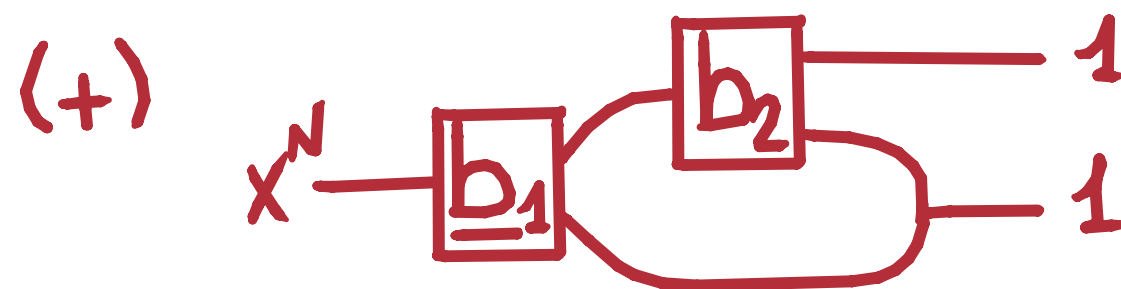
$\llbracket P \rrbracket \geq \llbracket c \rrbracket; \llbracket Q \rrbracket$

$$| b_1 +_b b_2$$

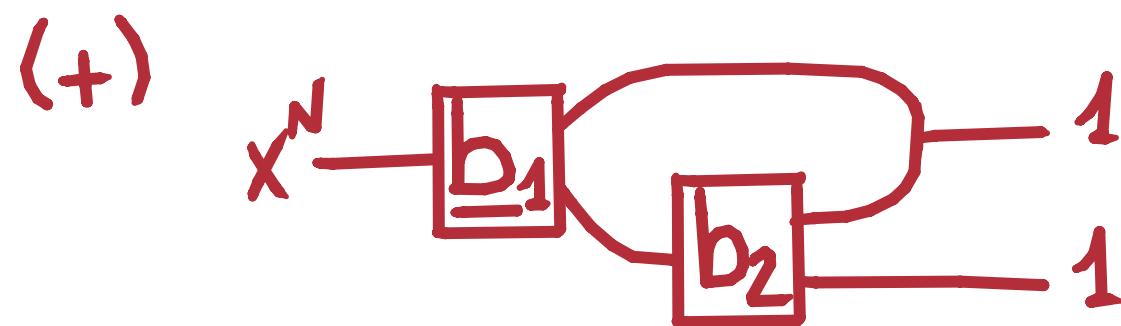


Conjunction and disjunction are derived:

$$b_1 \wedge b_2 := b_2 +_{b_1} R$$



$$b_1 \vee b_2 := L +_{b_1} b_2$$



THEOREM

The derivation rules for state and predicate (in)correctness triples are sound in any poset-enriched imperative category.

THEOREM

Commands and guards in an imperative category form a guarded monoid.

Commands with total and deterministic guards in a poset-enriched imperative category with least or greatest fixpoints satisfy all the axioms of GKAT ($U1-U5, S1-S5, W1-W2$) except for one: they satisfy a posetal version of the fixpoint axiom ($W3$).